

Multi-Modal, Training-Free Crack Extraction via Generalized FRANGI Graph

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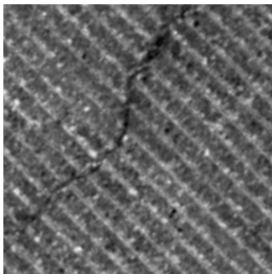
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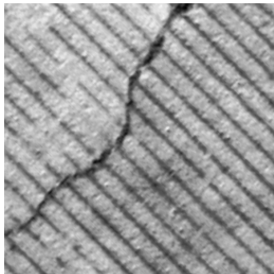
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Crack networks: a geometric problem



Intensity



Range / depth

Two aligned views of the same crack. [FIND 22]

Thin, weak and discontinuous.

Texture, illumination and sensors change the appearance.

Annotations are expensive.

Can we extract a useful network *without training*?

Our output: an editable crack skeleton.

| One or several scalar modalities; a shared **dark-ridge prior**.

[FIND 22] S. Zhou, C. Canchila, W. Song, FIND dataset, 2022. [EUVIP 26] L. Hauseux et al., this paper.

01

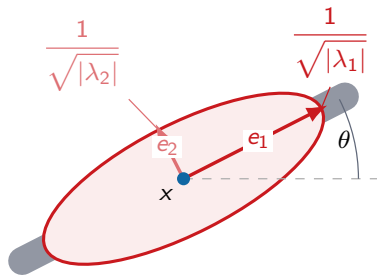
From pixels to a FRANGI graph





The Hessian describes local ridge geometry

$$\mathcal{H}_\sigma(x) = \nabla^2(G_\sigma * I)(x), \quad |\lambda_1| \leq |\lambda_2|.$$



Shape $R_B = |\lambda_1|/|\lambda_2|$

Near 0: elongated ridge. Near 1: blob.

Contrast $S = \sqrt{\lambda_1^2 + \lambda_2^2}$

Strength of the local curvature.

Direction e_1 runs along the ridge.

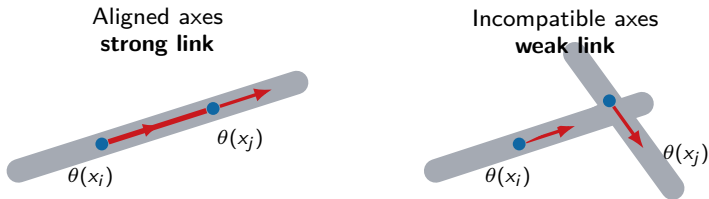
$$F_\sigma(x) = \exp\left(-\frac{R_B^2}{2\beta^2}\right) \left(1 - \exp\left(-\frac{S^2}{2c^2}\right)\right) \text{ if } \lambda_2 > 0; \text{ otherwise } 0.$$

Classical FRANGI: one score per pixel; retain the best scale.

[Frangi 98] A. F. Frangi et al., "Multiscale vessel enhancement filtering," MICCAI, 1998. Diagram adapted from the thesis defense.

What the graph adds: orientation alignment

The filter scores pixels separately; the graph checks **compatibility between pixels**.



$$S_{\text{align}}^{\sigma}(i, j) = \exp \left[-\frac{1}{2} \left(\frac{\sin(\theta^{\sigma}(x_i) - \theta^{\sigma}(x_j))}{s_a} \right)^2 \right]$$

The sine respects unoriented axes: θ and $\theta + \pi$ are equivalent.

$$S_{ij}^{(0)} = \max_{\sigma \in \Sigma} \underbrace{S_{\text{int}}^{\sigma} S_{\text{shape}}^{\sigma}}_{\text{curvature and elongation pairwise refinement}} \underbrace{S_{\text{align}}^{\sigma}}$$

[GRETSI 25; EUVIP 26] L. Hauseux et al. Curvature and elongation are refined by orientation compatibility. Diagram adapted from the defense.

Fuse modalities before measuring ridge geometry

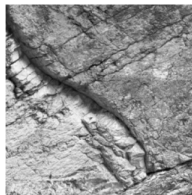
1. Normalize each Hessian

$$\hat{\mathcal{H}}_{\sigma}^{(m)}(x) = \frac{\mathcal{H}_{\sigma}^{(m)}(x)}{\max_{y \in \Omega} \|\mathcal{H}_{\sigma}^{(m)}(y)\|_2}.$$

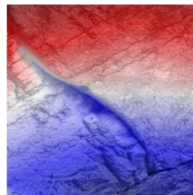
2. Fuse the normalized operators

$$\mathcal{H}_{\sigma}^{\text{fused}}(x) = \sum_m w_m \hat{\mathcal{H}}_{\sigma}^{(m)}(x).$$

$w_m \geq 0$, $\sum_m w_m = 1$. Normalization is per modality and per scale.



Intensity



Depth overlay

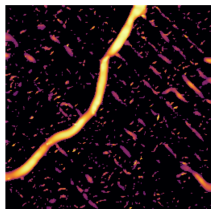
Aligned modalities share the crack geometry; sensor-specific artifacts may differ.

Palais des Papes, France. © Cerema.

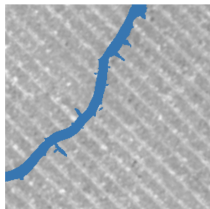
| Compute **eigenvalues and directions after fusion** — do not average FRANGI response maps.

[EUVIP 26] Section III-B. $\|\cdot\|_2$: spectral norm. Aligned modalities and consistent ridge polarity are required.

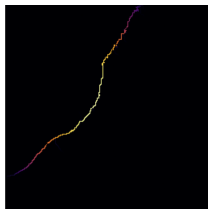
From pairwise similarities to an editable skeleton



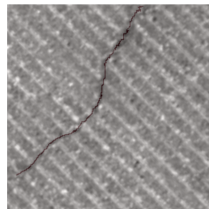
Edge similarity



Retained component



Tree centrality



Extracted skeleton



For $0 < \rho_{ij} = \|x_i - x_j\|_2 \leq R$:

$$d_{ij} = \min\{1, \rho_{ij}(1 - S_{ij}^{(0)})\}, \quad S_{ij} = 1 - d_{ij}.$$

Similarity S_{ij} for selection and centrality; **cost** d_{ij} for the tree.

[EUVIP 26] Sections III-C–D. FIND: retain the largest component (one target crack). Parameters are fixed per dataset.

02

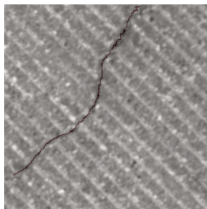
Experimental results



FIND: supervised learning wins on clean data



Ground truth



Our skeleton

An interpretable network is recovered without training annotations.

Clean FIND, intensity + range

Method	IoU $\uparrow$	Tversky $\uparrow$	W $\downarrow$
Our graph	60%	68%	18 px
CrackSegDiff	87%	90%	5 px

Our modality ablation: IoU

I	R	$I + R$	$I + R + F$
41%	54%	60%	63%

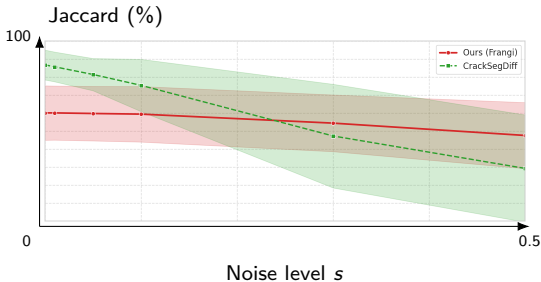
F is filtered range, not an independent sensor.

| A complementary, training-free method — **not a new clean-data winner.**

Evaluation: both skeletons dilated by 3 px for overlap; Tversky $(\alpha, \beta) = (1, 0.5)$.

[FIND 22; CrackSegDiff 24] [EUVIP 26] Tables II–III. IoU = Jaccard; W : Wasserstein skeleton distance.

FIND: slower degradation under synthetic noise



— Our graph — CrackSegDiff

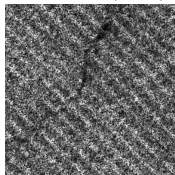
Shaded bands: standard deviation. Original paper figure.

| Lower initial accuracy, but a **more gradual decline** in this experiment.

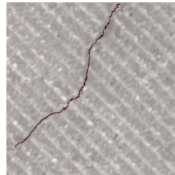
Controlled sensor perturbations

$$I_s = I(1 + N), \quad R_s = R + N',$$

$N, N' \sim \mathcal{N}(0, s^2)$, independent.



Noisy intensity

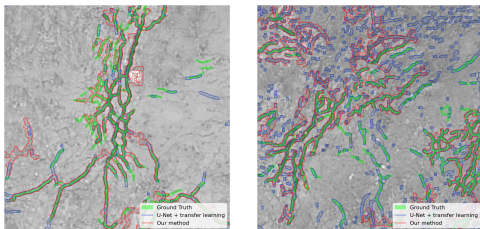


Our result, $s = 0.5$

[EUVIP 26] Section IV-C. Original noise-degradation figure. Results apply to the tested synthetic perturbations.

Geological images: transfer without retraining

Vaches Noires: visible UAV images



Test1

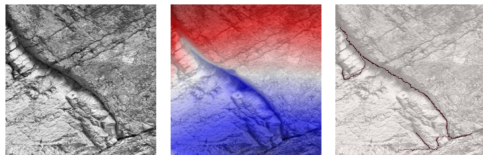
Test2

Ours Ground truth Adapted U-Net

IoU, ours / U-Net: 47 / 50%; 29 / 34%.

U-Net was adapted on a separate annotated image from the same area.

Palais des Papes: intensity + depth



Intensity

Depth overlay

Our skeleton

The main fracture is recovered.

Minor cracks are missed where contrast or depth resolution is insufficient.

No learned weights; parameters are chosen for the dataset.

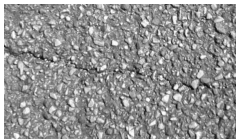
[EUVIP 26] Section IV-D. Images: © Cerema. Two annotated patches and one qualitative case: exploratory transfer evidence.

03

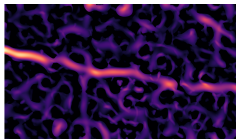
Beyond pairwise connectivity



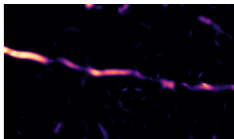
A granular background motivates order $K = 2$



Granular image



Pixel-wise FRANGI



Graph similarity



Retained component

Pebbles also produce ridge responses: a chain of links can connect the background.

$K = 2$: **connect edges through triangles**, not just through shared vertices.



$K = 1$



$K = 2$

The bare bridge no longer merges the two triangle-connected groups.

An exploratory extension, beyond the EUVIP evaluation.



Conclusion and perspectives

What we contribute

A FRANGI graph combining local ridge geometry with pairwise alignment.

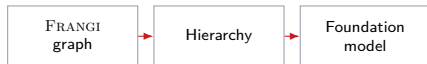
Operator-level multimodal fusion and an editable, training-free skeleton.

A complementary tool when annotations are scarce or inputs are degraded.

Where to go next

Validate higher-order connectivity and sensitivity on broader datasets.

Compare with SAM and its crack-specific adaptation, CrackSAM. [SAM 23; CrackSAM 24]



Can the FRANGI graph and its hierarchy guide a foundation model more effectively?

A research question — not a demonstrated gain.



Bibliography: method and experiments

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